

☒ Concept of Windows Server Backup (WSB)

Windows Server Backup (WSB) is a **built-in feature** of Microsoft Windows Server operating systems designed to help administrators **protect data and recover systems** after data loss, hardware failures, or disasters. It offers **basic to advanced backup and recovery functionalities** for files, system states, volumes, and full servers.

☒ 1. What is Windows Server Backup?

Windows Server Backup (WSB) is a feature that provides a **GUI-based and command-line interface** for backing up and restoring:

- Files and folders
- Volumes
- System state
- Entire server (bare-metal recovery)

Introduced in Windows Server 2008, it replaced the older NTBackup utility.

❏ 2. Features of Windows Server Backup

Feature	Description
Full Server Backup	Backs up all volumes, including system files and applications.
System State Backup	Backs up key system components like registry, boot files, Active Directory (if domain controller).
Volume Backup	Backs up specific disk volumes.
File/Folder Backup	Select specific files or folders.
Bare-Metal Recovery	Restore entire server to a new hardware setup.
Incremental Backup	Only backs up changes since the last backup, saving time and space.
Scheduled Backup	Automate backup tasks using scheduling.
VSS (Volume Shadow Copy Service)	Ensures backups of open/locked files are consistent.

☒ 3. Types of Backups in WSB

1.

Full Backup

- Backs up everything selected (including system).
- Required for the first backup.
- Used for disaster recovery.

2.

Incremental Backup

- Default in WSB after full backup.
- Uses VSS to track changes since the last backup.

3.

System State Backup

- Includes boot files, system registry, Active Directory database (if applicable), and COM+ class registration.

4.

Bare-Metal Backup

- Includes system state + enough data to restore to a clean drive or hardware.

☒ 4. Installation of WSB

WSB is not installed by default on Windows Server. To install:

Using Server Manager:

1.
Open Server Manager > **Manage** > **Add Roles and Features**
2.
Select **Windows Server Backup** under **Features**
3.
Click **Install**

Using PowerShell:

powershell

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```
Install-WindowsFeature -Name Windows-Server-Backup
```

> 5. Tools for WSB Management

1.

GUI Tool:

- Accessible via Server Manager → Tools → Windows Server Backup

2.

Command-Line Tool:

- wadmin.exe for managing backup via CLI and automation.

☒ 6. Backup Storage Options

- Local Disk
- External USB Drive

- Network Share (SMB Path)
 - DVD or removable media (*less common in modern environments*)
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☒ 7. Restoration Options in WSB

Scenario	Tool/Option
File/folder recovery	WSB GUI or wbadmin start recovery
System State recovery	Command line only (wbadmin start systemstaterecovery)
Bare-metal restore	Windows installation media ☒ Recovery options ☒ Load backup
Volume recovery	Can restore individual volumes without affecting others

☒ 8. Disaster Recovery with WSB

- Boot from Windows Server installation media.
 - Choose **Repair your computer** ☒ Troubleshoot ☒ **System Image Recovery**.
 - Select the latest backup to restore full server image.
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☒ 9. Security Considerations

- Backup files are not encrypted by default. Use **BitLocker** or store on secured network shares.
- Backup users need **Backup Operators** or **Administrators** group membership.
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Backups can be protected against ransomware by using WORM (Write Once Read Many) storage.

❏ 10. Limitations of WSB

Limitation	Details
Cannot backup to tape	Use third-party tools for tape backups.
One backup schedule per server	No per-volume or per-folder schedules.
Limited retention policy	Old backups overwritten unless managed manually.
No cloud support natively	Integration with Azure requires Azure Backup agent or third-party tools.

❏ 11. Common Commands with wbadmin

Command	Purpose
wbadmin start backup	Manually start a backup
wbadmin get versions	List available backups
wbadmin start recovery	Restore files or folders
wbadmin start systemstaterecovery	Restore system state
wbadmin delete systemstatebackup	Delete system state backups

❏ 12. Use Cases for Windows Server Backup

- Routine **data protection** for Windows Servers.
- Create **recovery images** before patching or configuration changes.
- Enable **quick disaster recovery** after ransomware or hardware failure.

- Backup and recovery for **domain controllers, file servers, application servers, etc.**